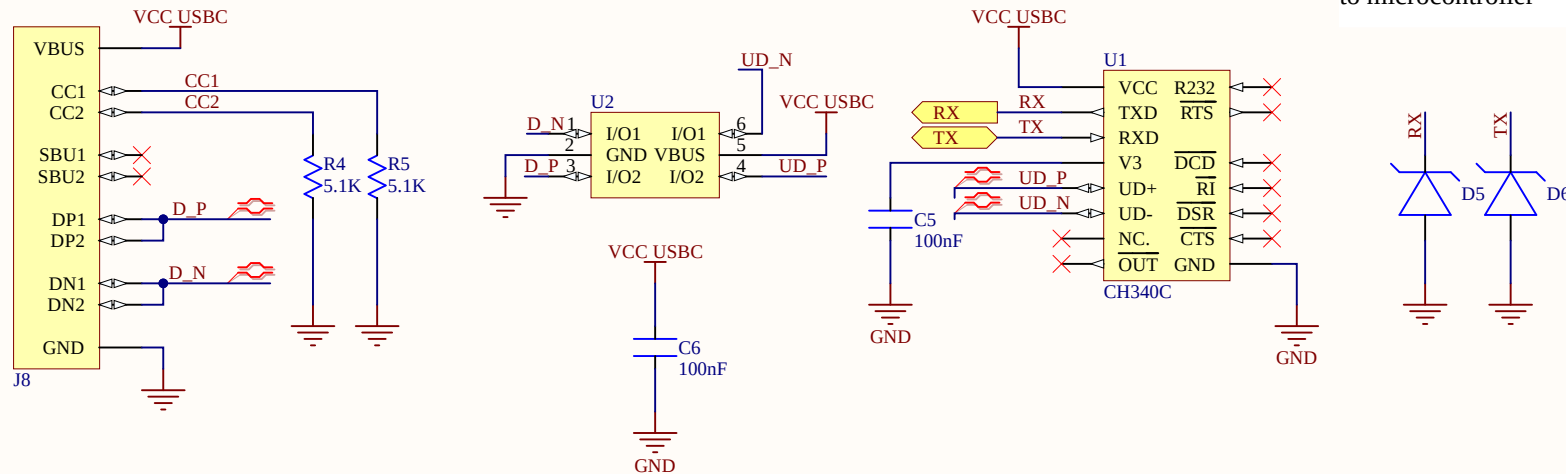
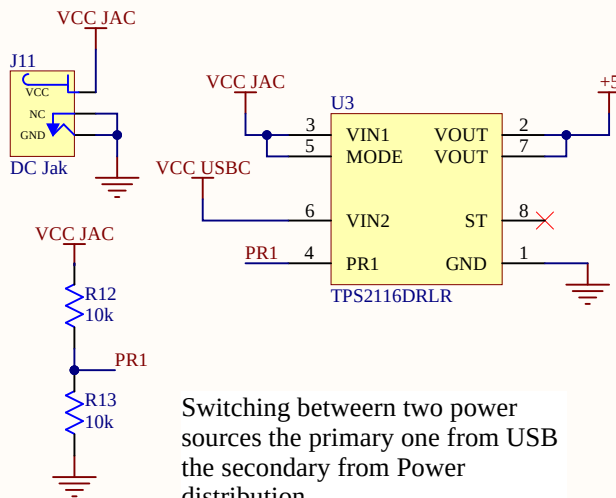


USB Connection

Signal here converted by USB to UART converter to be connected to microcontroller



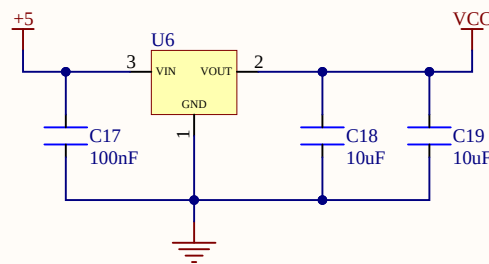
Power Switching



Switching between two power sources the primary one from USB the secondary from Power distribution

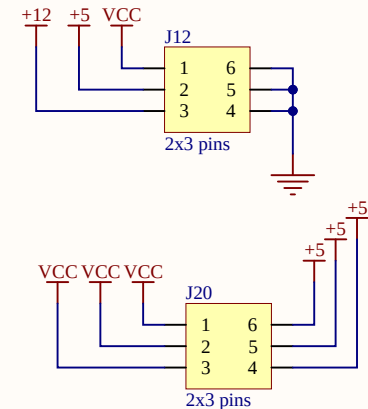
LDO

VCC => 3.3V



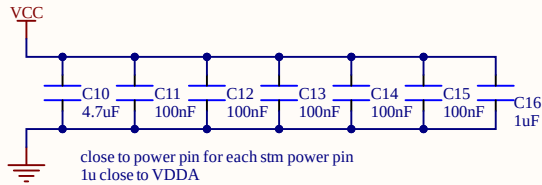
regulating working 5V to 3.3V

Additional power pins



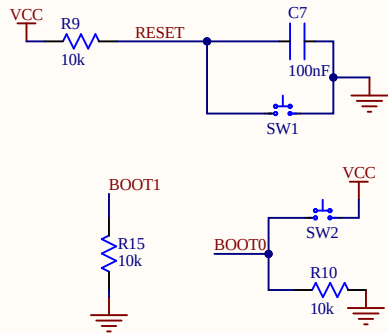
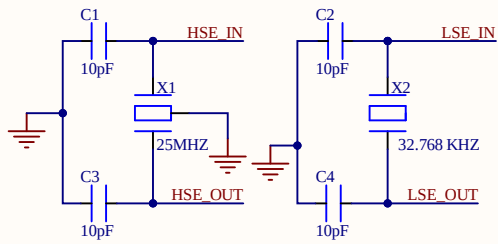
Title		
Power Management		
Size	Number	Revision
A		
Date:	8/29/2026	Sheet of
File:	D:\Control board\..POW.SchDoc	Drawn By:

power efficiency

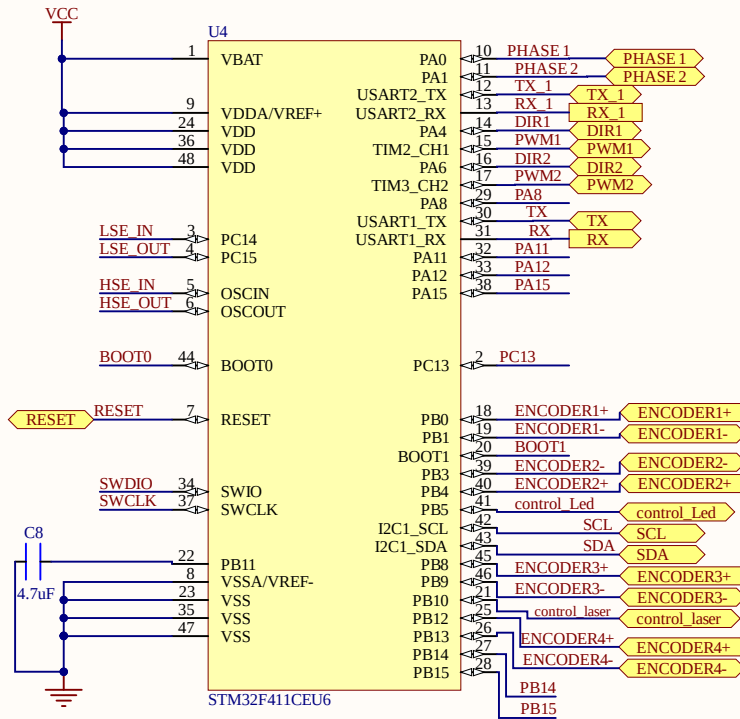


close to power pin for each stm power pin
1u close to VDDA
4.7u close to ic not specific pin
4.7u close to VCAP pin for coupling internal LDO

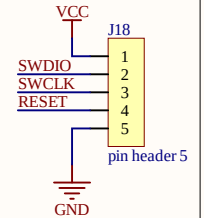
Crystals



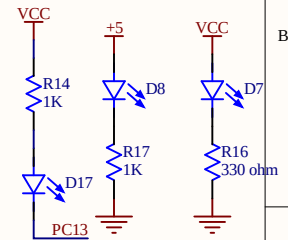
Main Controller



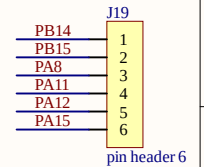
ST link



Power indicators

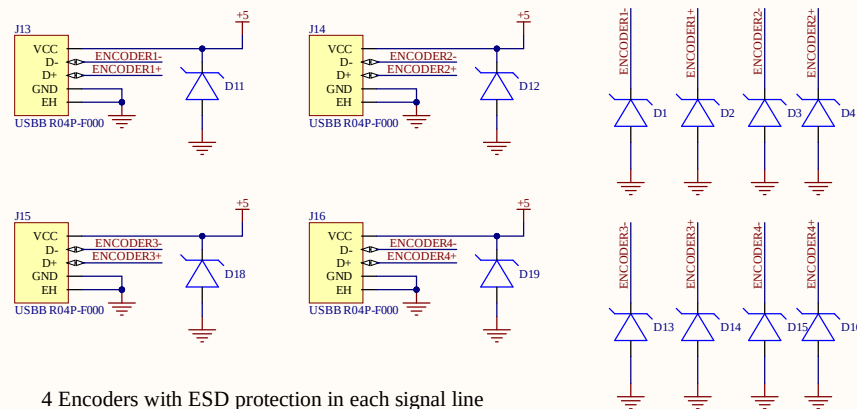


Additional Pins

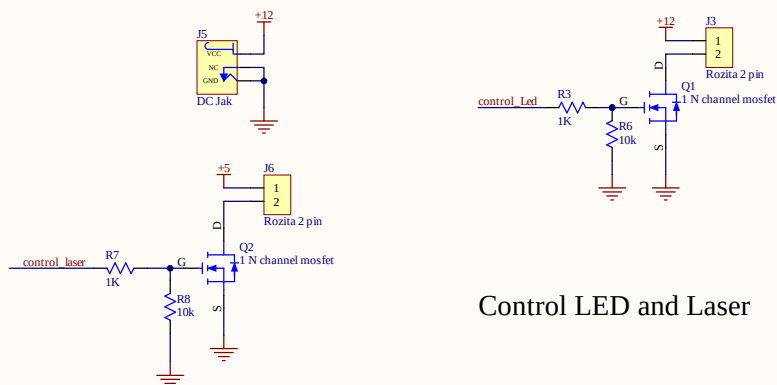


Title			STM32
Size	Number	Revision	
A4			
Date:	8/29/2026	Sheet of	
File:	D:\Control board\..MCU.SchDoc	Drawn By:	

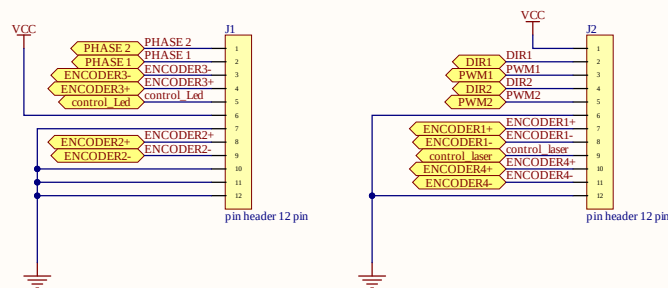
Encoders



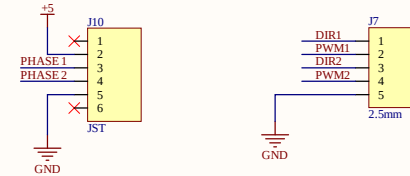
Control



For testing in case not connect in specific component in system

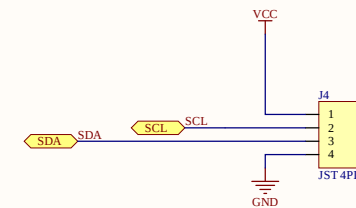


Driller Connection



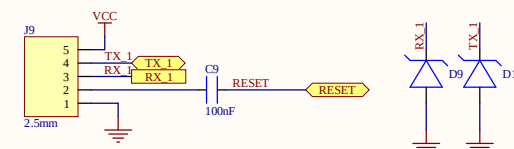
J7 connect to 2 DC motor & J10 connect to one encoder on one motor

Motion Control

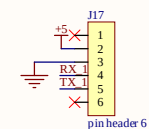


Connect to I2C Hub to control Motors of rover

Uart Connection



Connect to Bluetooth



Title Actuators & Sensors		
Size A3	Number	Revision
Date: 8/29/2026	Sheet of	
File: D:\Control board\Act & Sen.SchDoc	Drawn By:	

